

WASP-10b

R-1562

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_200828 · created 2026-08-01T21:41:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459089.7305 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.015
Transit depth (Rp/Rs)^2	2.82 +/- 0.49 [%]
Orbital Inclination (inc)	87.81 +/- 1.0
Airmass coefficient 1 (a1)	0.917 +/- 0.016
Airmass coefficient 2 (a2)	0.0749 +/- 0.008
Scatter in the residuals of the lightcurve fit is	1.8 %
Best Comparison Star	#2 - [380, 445]
Optimal Aperture	3.78
Optimal Annulus	10.75
Transit Duration (day)	0.0874 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.015 vs 0.1592 (deviation 0.59σ)
Duration fit vs published	0.0874 d vs 0.0946 d (deviation 1.0σ)
Mid-time O–C	-1.9 min
Depth SNR	11.2
Residual scatter	1.8 %

Light curve

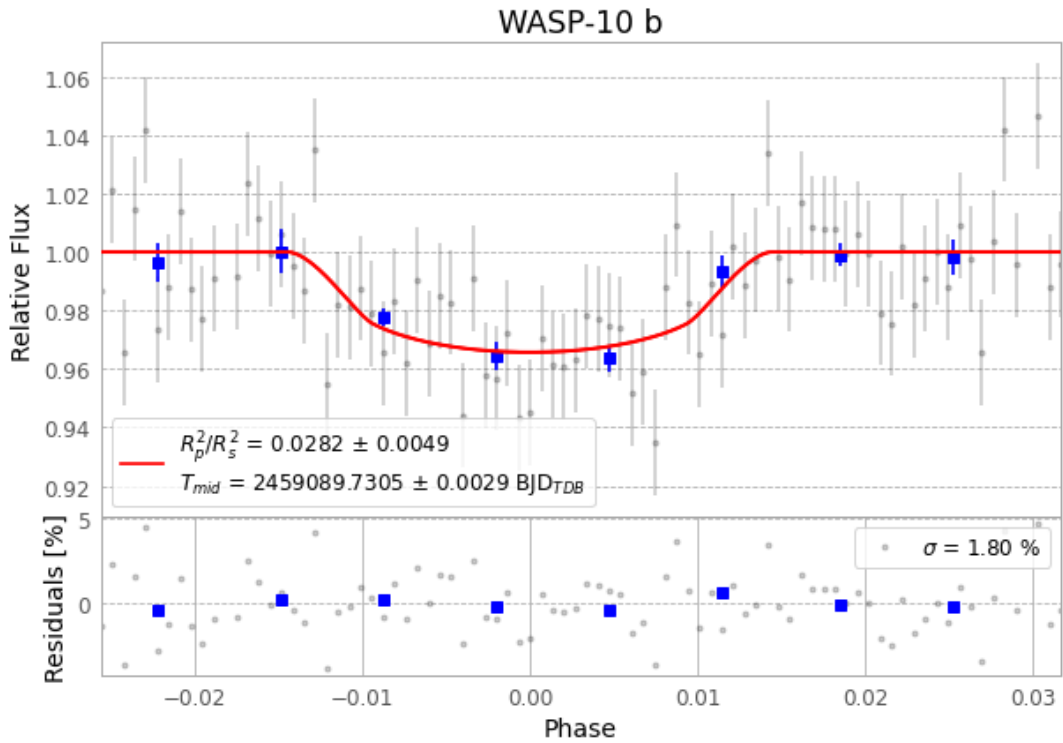


Figure 1 — FinalLightCurve_WASP-10 b_2020-08-27.png (SHA-256 b10f6cc9f75b1a6f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [307, 147], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01c45bf82e7d8ee4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), WASP-10200828033012.FITS → WASP-10200828074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-200828.log (a35113798ae4...), FinalLightCurve_WASP-10 b_2020-08-27.png (b10f6cc9f75b...), FinalLightCurve_WASP-10 b_2020-08-27.csv (2079c2564cdf...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:41Z.